Recitation 6

Math 311 Fall 2026
Rutgers University

Exercise 1 Let S be the set of positive integers whose decimal representations contain no digit 0. Define

$$s_n = \sum_{1 \le j \le n} \frac{1}{j}$$

for $j \in \mathbb{S}$.

(a) For $k \in \mathbb{N}$, let

$$S_k = \{j \in \mathbb{S} : 10^{k-1} \le j \le 10^k\}.$$

Then every digit of j has 9 choices. Since there are k digits for S_k, it follows that

$$|S_k| = 9^k.$$

By definition of S_k,

$$j \ge 10^{k-1},$$

so

$$\frac{1}{j} \le \frac{1}{10^{k-1}}.$$

Hence

$$\begin{aligned} \sum_{j \in S_k} \frac{1}{j} &\le \frac{|S_k|}{10^{k-1}} \\ &= 9\left(\frac{9}{10}\right)^{k-1}. \end{aligned}$$

(b) Let $\epsilon > 0$. Since

$$90(\frac{9}{10})^{k-1} \to 0,$$

we can choose $K \in \mathbb{N}$ such that

$$90(\frac{9}{10})^{K-1} < \epsilon.$$

Set $N = 10^{K-1}$, and let $m > n \geq N$. Then

$$\begin{aligned}
|s_m - s_n| &= \sum_{n<j\leq m} \frac{1}{j} \\
&\leq \sum_{k=K}^{\infty} \sum_{j\in S_k} \\
&\leq \sum_{k=K}^{\infty} 9\left(\frac{9}{10}\right)^{k-1} \\
&= 90\left(\frac{9}{10}\right)^{K-1} \\
&< \epsilon.
\end{aligned}$$

So s_n is cauchy. Thus, the Cauchy Criterion says that $s_n \to L$ for some $L \in \mathbb{R}$. Since $s_1 = 1$ and S only contains positive terms, it follows that $s_n \geq 1$ for all $n \in \mathbb{N}$. Thus by the Order Limit Theorem, $L \geq 1 > 0$. Next, from part (a) we have

$$\begin{aligned}
s_n &\leq \sum_{k=1}^{\infty} \sum_{j\in S_k} \frac{1}{j} \\
&\leq \sum_{k=1}^{\infty} 9\left(\frac{9}{10}\right)^{k-1} \\
&= \frac{9}{1-\frac{9}{10}} \\
&= 90.
\end{aligned}$$

Therefore, by the Order Limit Theorem

$$0 < L < 90.$$

Exercise 2

(a) Let $\{x_n\}_{n\in\mathbb{N}}$ be a real sequence, and define

$$u_n = x_n + x_{n+1}, \qquad v_n = x_n + x_{n+2}.$$

Suppose u_n and v_n are Cauchy. Notice that

$$\begin{aligned}
u_n + u_{n-1} - v_{n-1} &= (x_n + x_{n+1}) + (x_{n-1} + x_n) - (x_{n-1} + x_{n+1}) \\
&= 2x_n.
\end{aligned}$$

Let $\epsilon > 0$. Since u_n is cauchy,

$$m, n \geq N_1 \implies |u_m - u_n| < \frac{2\epsilon}{3}.$$

Similarly, since v_n is cauchy,

$$m, n \geq N_2 \implies |v_m - v_n| < \frac{2\epsilon}{3}.$$

Choose $N = \max\{N_1, N_2\} + 1$. Then

$$\begin{aligned}
|x_m - x_n| &= \frac{1}{2}|(u_m + u_{m-1} - v_{m-1}) - (u_n + u_{n-1} - v_{n-1})| \\
&\leq \frac{1}{2}\left(|u_m - u_n| + |u_{m-1} - u_{n-1}| + |v_{m-1} - v_{n-1}|\right) \\
&\leq \frac{1}{2}\left(\frac{2\epsilon}{3} + \frac{2\epsilon}{3} + \frac{2\epsilon}{3}\right) \\
&= \epsilon.
\end{aligned}$$

Therefore x_n is cauchy.

(b) Suppose $\sum_{n=1}^{\infty} a_n^2$ and $\sum_{n=1}^{\infty} b_n^2$ both converge. Notice that

$$\begin{aligned}
(|a_n| - |b_n|)^2 &\geq 0 \\
a_n^2 - 2|a_n||b_n| + b_n^2 &\geq 0 \\
\frac{a_n^2 + b_n^2}{2} &\geq |a_n||b_n|.
\end{aligned}$$

Next, by the Algebraic Limit Theorem, we have that

$$\sum_{n=1}^{\infty} \frac{a_n^2 + b_n^2}{2} \quad \text{converges.}$$

Moreover, by the comparison test

$$\sum_{n=1}^{\infty} |a_n||b_n| \quad \text{converges.}$$

Hence $\sum_{n=1}^{\infty} a_n b_n$ converges absolutely. Then, by theorem 2.7.10, we can conclude that ever rearrangement of this series also converges.

Exercise 3 For $m, n \in \mathbb{N}$, define

$$a_{m,n} = \frac{mn^2}{m^2 + n^4}.$$

(a) For $\lim_{m\to\infty} \lim_{n\to\infty} a_{m,n}$, we first divide through by n^4 and then take $n \to \infty$,

$$\lim_{n\to\infty} a_{m,n} = \lim_{n\to\infty} \frac{\frac{m}{n^2}}{\frac{m^2}{n^4} + 1} = 0.$$

Taking $m \to \infty$ does not do anything, so $\lim_{m\to\infty} \lim_{n\to\infty} a_{m,n} = 0$.

For $\lim_{n\to\infty} \lim_{m\to\infty} a_{m,n}$, we first divide through by m^2 and then take $m \to \infty$,

$$\lim_{m\to\infty} a_{m,n} = \lim_{n\to\infty} \frac{\frac{n^2}{m}}{1 + \frac{n^4}{m^2}} = 0.$$

Taking $n \to \infty$ does not do anything, so $\lim_{n\to\infty} \lim_{m\to\infty} a_{m,n} = 0$.

(b) No, consider $m = n^2$. Then

$$\begin{aligned} a_{n^2,n} &= \frac{n^2 \cdot n^2}{(n^2)^2 + n^4} \\ &= \frac{n^4}{2n^4} \\ &= \frac{1}{2}. \end{aligned}$$

Hence,

$$\lim_{n\to\infty} a_{n^2,n} = \frac{1}{2} \neq 0.$$

(c) Suppose that $a_{m,n} \to L$ as $m, n \to \infty$. Because $a_{n,n} \to 0$, the uniqueness of limits implies that $L = 0$. Moreover, convergence of the double sequence to L forces its diagonal sequence to converge to L. Now choose

$$\boxed{\varepsilon_0 = \frac{1}{4}}.$$

Let $N \in \mathbb{N}$ be arbitrary. Choose $n \geq N$ and set $m = n^2$. Since $n \geq 1$,

$$m = n^2 \geq n \geq N.$$

But

$$\begin{aligned} |a_{m,n} - L| &= |a_{n^2,n} - 0| \\ &= \left|\frac{1}{2}\right| \\ &= \frac{1}{2} \\ &\geq \frac{1}{4} = \varepsilon_0. \end{aligned}$$

Thus, for every $N \in \mathbb{N}$, there exist $m, n \geq N$ satisfying

$$|a_{m,n} - L| \geq \varepsilon_0.$$

This contradicts the definition of double-sequence convergence. Therefore,

$$\{a_{m,n}\}_{m,n\in\mathbb{N}} \text{ does not converge as } m, n \to \infty.$$